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| **PI Name:** | Koch, Donald Ph.D. | | |
| **Project Title:** | Dynamic Response of Particle-Filled Polymeric Fluids in Flows with Complex Lagrangian Time Histories | | |
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| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Complex fluids | | |
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| **Human Research Program Risks:** | None | | |
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| | Particle-filled polymeric fluids, frequently encountered in processing composite materials into fibers or sheets, turbulent drag reduction, and hydraulic fracturing, undergo alternating periods of simple shear and extensional motion in a Lagrangian reference frame. The dynamic rheological response of a particle-polymer mixture is not a simple superposition of independent polymeric and particle stresses. Instead, there is significant coupling between the velocity disturbance produced by the particles, polymer stress, and particle orientation. There is a dearth of relevant experiments and theoretical/ numerical analyses exploring the dynamic rheology of these systems. The proposed project provides an opportunity to employ theoretical and numerical simulation methods uniquely suited to predict the particle-induced polymer stress in transient flows and validate these methods by comparison with transient extensional rheology experiments of particle-polymer mixtures subject to pre-shearing in the Physical Sciences Informatics (PSI) database. | | |

**Task Description:**

A filament stretching rheometer used in the Shear History Extensional Rheology Experiments (SHERE, SHERE II, SHERE-Reflight / SHERE-R) enabled the measurement of a particle-filled polymeric fluid's transient extensional flow response subject to variable pre-shearing rates. SHERE and SHERE-R provide complementary measurements for the same fluid without particles. The experiments use dilute particle suspensions in a well-characterized Boger fluid over a range of moderate extension rates, conditions unachievable in Earth's gravity. A statistical analysis of the SHERE data indicates a significant reduction in strain hardening by particles at high strains, which agrees with a preliminary theoretical analysis.

Prediction of the coupled particle and polymer stresses is arduous because the particle-induced velocity disturbance greatly alters the polymer stress in regions close to the particle surface, leading to strong polymer stress gradients that are difficult to resolve numerically and subtle to predict theoretically. The interplay between particle orientation, velocity disturbance, and polymer stresses when fibers are suspended in polymeric fluids presents further challenges.

The project will employ theory and numerical simulations developed in our group, uniquely suited to address these problems. A finite difference method will solve the inertia-less viscoelastic flow in a spheroidal coordinate system that conforms to and rotates with the surface of a spheroidal particle and applies a time-varying linear velocity boundary condition at a nearly spherical outer boundary far from the particle. A demonstration that the ensemble average of the linear perturbation to the polymer stress due to the particle velocity disturbance is zero is essential to obtain convergent bulk rheological properties from numerical simulations.

A complementary theoretical approach will yield semi-analytical predictions of the suspension rheology valid in the limit of small polymer concentration, c. Comparison with the numerical simulation will reveal the range of c, over which this theory is quantitatively accurate, and provide useful qualitative predictions for larger c. The theory is less computationally intensive and avoids computational stability limitations typical of nonlinear viscoelastic flow computations at high Deborah numbers and polymer extensibilities. Hence, it facilitates exploring broader parameter regimes.

These methods will be applied to study the rheology of (1) suspensions of spheres in transient extensional flow with and without preshearing for comparison with the SHERE experiments; (2) suspensions of spheroids experiencing a simple shear flow preceded and followed by a period of extensional flow to model the processing of fiber-filled polymeric films; and (3) suspensions of spheroids subject to simple shear flow alternating with short bursts of extensional motion as a model of turbulent drag reduction by the coupled effects of fiber and polymer additives.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Suspensions of particles within polymeric or viscoelastic fluids are integral to numerous manufacturing and industrial processes. Within these applications, these suspensions navigate intricate flows, ranging from shear caused by solid surfaces to extension through orifices. The purpose behind introducing particles, such as spheres or fibers, is often to confer specific attributes upon the final product, such as heightened mechanical strength or directional electrical conductivity. However, the interaction between particles and polymers within the fluid is a pivotal factor dictating the energy demands throughout the manufacturing process. This interaction significantly alters the rheological properties of the suspension in comparison to the particle-free polymeric fluid. This study provides quantitative predictions and mechanistic understanding of the effects of particle-polymer interactions on the flow resistance of such materials.

Industrial applications such as hydraulic fracturing, extrusion molding, fiber spinning, roll-to-roll manufacturing of thin films for flexible and transparent electronics, and oil recovery rely on polymeric fluids laden with particulate matter. Particles are added for various purposes, such as improving anisotropy in the mechanical or thermal properties of the manufactured flexible film, imparting strength to the extruded product, or acting as proppants to keep the fractured rock open in hydraulic fracturing. In fiber spinning, the particle-filled fluid is first sheared within the spinneret and then pulled by the drawing mechanism in the jet, leading to a strong uniaxial extensional flow before solidifying to form a fiber. In hydraulic fracturing, the fluid first undergoes strong extensional flow at the entrance to a narrow pore and is then sheared while flowing through the pore. In roll-to-roll manufacturing, the particle-filled viscoelastic liquid is injected into the patterning channel, where it experiences extension, followed by shear in the channel and a second period of extension during film formation. In this report, we summarize our findings on the effect of particle-polymer interaction and its influence on particle dynamics and suspension rheology in a variety of cases inspired by the complex Lagrangian time histories experienced by this system in the aforementioned applications.
The flow around each particle in the aforementioned studies is locally linear, and large gradients in polymer stress are expected, particularly as the concentration or relaxation time of the polymers is increased. This has prevented previous studies from exploring the industrially relevant parameter space. Therefore, we develop a numerical method [1] based on prolate spheroidal coordinates that exactly models the particle surface as one of the coordinate surfaces. This method allows us to span an extensive parameter space and discover novel mechanisms. [Ed. Note: For citations, see the list of publications below and the Cumulative Bibliography.]

Understanding the extensional rheology of viscoelastic suspensions is crucial for optimizing the efficiency of the industrial processes discussed above, but remains less explored compared to shear rheology, even for the simplest particle shape, i.e., sphere. Our studies [2,3] explore the extensional rheology of dilute sphere-filled polymeric liquids for a broad range of polymer concentration, relaxation times, and the duration for which the uniaxial extensional flow is applied. From this comprehensive parameter study, we discover novel physical mechanisms and offer a qualitative comparison experiments performed in the SHERE investigations on the International Space Station. As the flows in practical applications are temporally varying, we investigate the rheology of sphere suspensions undergoing oscillatory shear and planar extensional flow. Our final investigations with spheres are conducted using simulations with mixed flows that are a sequence of extensional flows with simple shear in different directions. The results on transient flows with spheres were recently presented at the American Physical Society's Division of Fluid Mechanics annual meeting [4].

To achieve anisotropy in applications such as roll-to-roll manufacturing mentioned above, anisotropic particles are used. Not only is the rheology—and hence the energy requirements—during the manufacturing process dependent on particle orientation in these cases, but the orientation is difficult to control passively if there is an inherent rotation in the imposed flow. Therefore, we began our investigation on spheroids with a study of orientation dynamics before providing rheological results. The orientation dynamics were investigated using a combination of slender body theory

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| **Task Progress:** | and numerical simulations, with the former recently published in [5].<br><br>Motivated by the prevalence of Lagrangian unsteady flows in industrial applications, we used our finite difference numerical solver to study polymeric flows around particles when the imposed flow field far from the particle varies with time. The results were employed to determine the transient rheological response of a dilute suspension of particles in a polymeric fluid. We first considered large amplitude oscillatory shear flow and determined the limiting strain amplitude above which the linear elastic Maxwell model is no longer applicable. While a particle-free fluid shear thins with increasing frequency and amplitude of oscillation, the particle-induced polymer stresses (PIPS) can lead to shear thickening of the intrinsic viscosity contributed by the particles.<br><br>A study of time periodic extensional flow is relevant to flows in porous media such as flows through arrays of posts in microfluidic channels, where the fluid flows through successive pore throats. In oscillatory planar extension, the addition of particles can lead to a decrease in the storage modulus for high amplitudes and moderate frequencies. This is driven by negative values of the polymeric contributions to the particles stresslet and the PIPS. This shows that the addition of particles can reduce the elastic stresses in unsteady extensional flows.<br><br>Another type of transient flow often encountered in material processing involves successive periods of simple shear and extension. We studied three such flows. First, we analyzed the impact of pre-shear applied perpendicular to the extensional axis, which is similar to the pre-shear experiments done as a part of the SHERE studies performed in micro-gravity on the International Space Station. Similar to their observations, we found that applying a pre-shear before extension can reduce strain hardening slightly, due to pre-alignment of polymers perpendicular to the extensional axis, but the overall effect is small. In contrast, we found that applying pre-shear along the extensional axis has a more significant impact. Pre-alignment of polymers along the extensional axis increases the stresses at the onset of the extensional motion. However, the peak stress contributed by the particle-polymer interactions as the strain amplitude develops is significantly reduced by the pre-shear. Furthermore, the addition of particles surprisingly reduced the stress of the material at large times. As a result, the addition of particles can limit the stress developed throughout the extensional flow following a longitudinal simple shear flow. This flow configuration is relevant to fiber spinning where the material experiences shear flow in a capillary before being stretched to form a thin fiber. We also examined the effect of applying extensional flow along the flow direction before shear flow, and find that, while pre-extension can reduce the shear stresses in a pure polymeric fluid at small time, the addition of particles does not enhance this effect.<br><br>Publications resulting from NASA grant 80NSSC23K0348:<br><br>[1] Sharma, A. and Koch, D.L. Finite difference method in prolate spheroidal coordinates for freely suspended spheroidal particles in linear flows of viscous and viscoelastic fluids. Journal of Computational Physics, 495:112559, 2023.<br><br>[2] Sharma, A. and Koch, D.L. Steady-state extensional rheology of a dilute suspension of spheres in a dilute polymer solution. Physical Review Fluids, 8(3):033303, 2023.<br><br>[3] Sharma, A. and Koch, D.L. Extensional rheology of dilute suspensions of spheres in polymeric liquids. Journal of Fluid Mechanics (submitted), 2025.<br><br>[4] Mukherjee, A., Sharma, A., Koch, D.L. Transient rheology of particle-filled polymer solutions. In APS Division of Fluid Dynamics Meeting Abstracts, pages L25–009, 2024.<br><br>[5] Sharma, A. and Koch, D.L. Rotation of a fibre in simple shear flow of a dilute polymer solution. Journal of Fluid Mechanics, 976:A9, 2023. |
| **Bibliography Type:** | Description: (Last Updated: 10/15/2025) |
| **Abstracts for Journals and Proceedings** | Mukherjee A, Sharma A, Koch D. "Transient rheology of particle-filled polymer solutions." 77th Annual Meeting of the APS Division of Fluid Dynamics, Salt Lake City, Utah, November 24-26, 2024.<br>Bulletin of the American Physical Society. 77th Annual Meeting of the APS Division of Fluid Dynamics, Salt Lake City, Utah, November 24-26, 2024. Abstract: L25.009. , Nov-2024 |
| **Articles in Peer-reviewed Journals** | Sharma A, Koch DL. "Extensional rheology of dilute suspensions of spheres in polymeric liquids." Journal of Fluid Mechanics. 2025 Sep 5;1018:A40. http://dx.doi.org/10.1017/jfm.2025.10530 , Sep-2025 |